TuneWorks Data Analysis

Independent Technical Report



"Helping clients make a change that matters..."

DSO552-D Group 7:

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Introduction

This report presents the independent technical analysis conducted for TuneWorks, a mid-sized entertainment agency seeking to better understand its operational performance through data. Using the provided dataset—which includes detailed records of engagements, customers, agents, and entertainers—we performed a series of SQL-based analyses to uncover patterns, trends, and strategic opportunities. Our goal was to explore the data systematically, evaluate its quality, calculate key business metrics, and generate insights that can inform decision-making around revenue, talent management, customer value, agent productivity, and overall operational efficiency. This document serves as a transparent research log of our process, including data exploration steps, analytical queries, observations, recommendations, and supporting visualizations.

Data Dictionary

Table 1: agents

Column Name	Data Type	Definition	Business Meaning
agentid	integer (key)	Unique identifier for each agent.	A unique ID number that connects each agent to the booking they made in the engagements table.
agtfirstname	varchar(20)	The agent's first name.	The agent's first name used to identify them.
agtlastname	varchar(20)	The agent's last name.	The agent's last name used to identify them.
agtstreetaddress	string	The street of the agent's address.	The street on which the agent lives.
agtcity	string	The city of the agent's address.	The city in which the agent lives.
agtstate	string	The state of the agent's address.	The state in which the agent lives.
agtzipcode	integer	The zipcode of the agent's address.	The specific location of the agent.
agtphonenumber	string	The agent's phone number.	The agent's work phone number through which to contact them.
datehired	date	Year-Month-Day of the date the agent was hired.	The date the agent was hired to work for TuneWorks to track how long the agent has been working in the company.
salary	integer	The agent's salary (yearly pay).	The base amount of money the agent makes in a year.
commissionrate	integer (decimal)	A percentage in decimal form.	The percentage of the booking contract price that an agent receives for each engagement, earned on top of their salary.

Table 2: customers

Column Name	Data Type	Definition	Business Meaning
customerid	integer(key)	A unique identifier assigned to each customer.	Each customer has a distinct ID. Used as the primary key for this table.
custfirstname	string	Customer's first name.	The customer's first name used to identify them.
custlastname	varchar	Customer's last name.	The customer's last name used to identify them.
custstreetaddress	varchar	The street address for each customer.	The street on which the customer lives.
custcity	varchar	The city of the customer's address.	The city in which the customer lives.
custstate	varchar	The state of the agent's address.	The state in which the customer lives.
custzipcode	varchar	The zip code of the agent's address.	The specific location of the customer.
custphonenumber	varchar	The customer's phone number.	The customer's phone number through which to contact them.

Table 3: engagements

Column Name	Data Type	Definition	Business Meaning
engagementnumber	integer	A unique identifier assigned to each engagement record. Not necessarily sequential.	Each engagement has a distinct ID. Used as the primary key for this table.
startdate	date	The calendar date when an engagement begins.	Represents the start day of the booking.
starttime	time	The exact start time of the engagement on the start date.	Combined with startdate, it defines the precise start timestamp of an event.

stopdate	date	The calendar date when an engagement ends.	Represents the end day of the booking.
stoptime	time	The exact end time of the engagement on the stop date	Combined with stopdate, it defines the precise end timestamp of an event.
contractprice	integer	The price for the engagement	The revenue earned from each engagement.
customerid	integer	A unique identifier assigned to each customer	Identifies the customer who booked the engagement. Foreign key to the customer table
agentid	integer	A unique identifier assigned to each agent	Identifies the responsible booking agent. Foreign key to the agents table
entertainerid	integer	A unique identifier assigned to each entertainer	Identifies which performer was booked. Foreign key to the entertainers table

Table 4: entertainer_members

Column Name	Data Type	Definition	Business Meaning
entertainerid	integer	A unique identifier referencing the entertainer record this member belongs to.	Connects a member to a specific entertainer.
memberid	integer	A unique identifier referencing the individual member record.	Identifies which member is associated with the entertainer.
Status	smallinteger	A numeric code indicating the member's current status within the entertainer's group (1 = active, 2 = inactive)	Check whether the member is active or not.

Table 5: entertainer_styles

Column Name	Data Type	Definition	Business Meaning
entertainerid	integer	Unique identifier for each entertainer.	Links to the Entertainers table. Identifies which entertainer the style belongs to.
styleid	integer	Unique identifier for each performance style or genre.	Links to the musical_styles table. Indicates the type of performance.
stylestrength	integer	Numeric scale indicating how strongly the entertainer performs in this style (1–3).	Reflects entertainer's proficiency level or expertise in that style. Useful for customer matching.

Table 6: Entertainers

Column Name	Data Type	Definition	Business Meaning
Entertainerid	Integer	Unique identifier in system for each entertainer	Serves as the primary key for the entertainers table. Links back to the engagements, entertainer_styles, and entertainer_members tables.
Entstagename	String	The name of each entertainer	Provides the unique name for entertainers and also brands the performers
Entssn	Integer	Government social security number	This is for tax implications for both the agent and the entertainer. These are not individual SSNs. These are for the entities.
Entstreetaddress	String	Physical mailing address for the	Allows the user to see the physical address to

		entertainers	mail business cards / tax information to
Entcity	String	Residing city of the entertainer	Identifies the residing cities
Entstate	String	Residing state of the entertainer	Identifies the residing state
Entzipcode	Integer	Residing zip code of the entertainer	Identifies the zip code
Entphonenumber	Integer	Phone number for entertainers	Primary contact numbers to get a hold of the entertainers
Entwebpage	String	Website for entertainers	Links to advertise events, promote, and brand the entertainers on the web
Entemailaddress	String	Email address for entertainers	Primary email address to get in contact with the entertainers
Dateentered	Date	The date that the data was entered in the entertainers table	Serves as the entry date for changes made to the entertainers' details

Table 7: members

Column Name	Data Type	Definition	Business Meaning
memberid	integer	A unique identifier assigned to each individual member record.	Primary key for the members table.
mbrfirstname	String	Member's first name.	Used to identify the member.
mbrlastname	String	Member's last name.	Used to identify the member.
mbrphonenumber	String	Member's phone number.	Used to communicate with the member.
gender	String	Member's gender.	Allows demographic or diversity-based analysis.

Table 8: Musical Preferences

Column Name	Data Type	Definition	Business Meaning
Customerid	int	A unique numeric identifier assigned to each client who wants to book entertainment	Links customers to this table to indicate their ranked preferences for entertainment styles.
Styleid	int	A unique numeric identifier that represents a genre from the music styles table	Shows the music styles that customers want most
Preferencesq	int	A numeric value of 1, 2, or 3 to rank the preferred styles of entertainment for each customer	Can be used by agents to manage entertainment bookings. In scenarios where double bookings can be an issue, ranked preferences can determine which customers value the entertainment more, while providing the agent with backup options.

Table 9: musical_styles

Column Name	Data Type	Definition	Business Meaning
styleid	integer	A unique numeric identifier assigned to each musical style record.	Serves as the primary key for this table and is referenced by entertainer_style. Distinguishes one musical style from another.
stylename	varchar	The name of the musical style or genre.	Represents the type of music that entertainers can perform and customers can request when booking.

Table 10: ztbIDays

Column Name	Data Type	Definition	Business Meaning
datefield	date	A single calendar date value for each row.	A list of consecutive dates used by the system for time calculations, testing, or joining with other tables when analyzing data by day.

Table 11: ztbISkipLable

Column Name	Data Type	Definition	Business Meaning
labelcount	integer	A simple numeric value that increases by 1 for each record.	Used by the systems to keep a running count or to label rows in order. This table does not store any customer, entertainer, or engagement info.

Table 12: ztbIWeeks

Column Name	Data Type	Definition	Business Meaning
weekstart	date	The first day of the week.	Marks when each week begins. Used for grouping or summarizing data by week.
weekend	date	The last day of the week.	Marks when each week ends. Used for grouping or summarizing data by week.

Table 13: ztblMonths

Column Name	Data Type	Definition	Business Meaning
monthyear	varchar	The month name combined with the year (January 2017).	Used to display or group information by month and year.
yearnumber	integer	The four digit year (2017, 2018).	Helps filter or summarize data by

			year.
monthnumber	integer	The month as a number from 1 to 12.	Ensures months can be sorted correctly in reports and charts.
monthstart	date	The first calendar day of the month.	Defines when each month begins; used for time-based joins or comparisons.
monthend	date	The last calendar day of the month.	Defines when each month ends; used in date range calculations.
january	integer	Indicates whether the month is January (1 = yes; 0 = no).	Allows quick filtering for January records.
february	integer	Indicates whether the month is February.	Allows quick filtering for February records.
march	integer	Indicates whether the month is March.	Allows quick filtering for March records.
april	integer	Indicates whether the month is April.	Allows quick filtering for April records.
may	integer	Indicates whether the month is May.	Allows quick filtering for May records.
june	integer	Indicates whether the month is June.	Allows quick filtering for June records.
july	integer	Indicates whether the month is July.	Allows quick filtering for July records.
august	integer	Indicates whether the month is August.	Allows quick filtering for August records.
september	integer	Indicates whether the month is September.	Allows quick filtering for September records.
october	integer	Indicates whether the month is October.	Allows quick filtering for October records.
november	integer	Indicates whether the month is November.	Allows quick filtering for November records.
december	integer	Indicates whether the month is December.	Allows quick filtering for December records.

Data Quality Notes

During the initial exploration of the TuneWorks database, several data quality considerations were identified across the core tables (engagements, agents, customers, entertainers, styles, and auxiliary system tables). These issues do not prevent analysis, but they influence interpretation, accuracy, and the reliability of downstream insights. The following summarizes the most important observations:

3.1 Missing or Non-Sequential Identifiers

- The engagementnumber column in the engagements table is not consecutive, suggesting that some historical records may have been removed, skipped, or were never captured.

- Similar primary keys (such as customerid, memberid, and entertainerid) appear clean and unique, but no system-level constraints are shown to confirm referential completeness.
  Impact: Analyses depending on engagement history (e.g., time-between-engagements) may slightly misrepresent true customer behavior.

3.2 Date and Time Inconsistencies

- Engagement start and stop timestamps are stored in separate date and time columns. This creates potential errors when engagements cross midnight or span multiple days, which would lead to negative or misleading durations if not combined correctly.

- A manual merge of (date + time) was required to compute accurate durations.
  Impact: Duration-based metrics (avg hours, revenue per hour, monthly duration trends) must be calculated carefully to avoid incorrect results.

3.3 Extreme or Outlier Engagement Durations

- One engagement in March spans approximately 218 hours, which is extremely long relative to the rest of the dataset.

- This single outlier heavily skews monthly duration averages.
  Impact: Monthly comparisons and duration vs. revenue relationships should either remove this outlier or interpret results accordingly.

3.4 Potential Redundancies and Incomplete Attributes in Address Fields

- Address fields across agents, customers, and entertainers are split into multiple columns (street, city, state, zip), which is good practice.

- However, there is no validation or standardized formatting (e.g., inconsistent capitalization, string vs. integer zip codes).
  Impact: Geographic analysis or merging by ZIP code may require data cleaning to ensure

consistency.

3.5 Possible Data Entry Errors in Demographic or Contact Attributes

- Phone numbers for agents, customers, and members are stored as free-form strings.
- Formats may be inconsistent (e.g., dashes, parentheses, missing area codes).
- Customer state and ZIP code definitions incorrectly reference “agent” in the description, which may be a documentation inconsistency rather than a data issue.
  Impact: Phone-based segmentation or spatial clustering may require additional cleaning.

3.6 Status & Strength Fields Are Encoded Without Descriptive Labels

- In entertainer_members, status is stored as numeric codes (1 = active, 2 = inactive) with no referential lookup table.
- In entertainer_style, stylestrength is stored as integers (1–3) without a defined scale beyond “strength.”
  Impact: Any analysis on activity status or strength needs proper decoding to avoid misinterpretation.

3.7 Auxiliary System Tables Are Not Meaningful for Business Analysis

- Tables such as ztblDays, ztblWeeks, and ztblSkipLabel appear to be system-generated or used for internal indexing.
- They contain no business attributes and should not influence primary analysis unless explicitly needed for time-series joins.
  Impact: These tables should be excluded from most reporting unless required for date alignment.

3.8 No Nullability or Missing Value Indicators Provided

- The provided schema does not specify whether fields allow NULL values.
- Some tables (e.g., members, entertainers) may have incomplete contact information or optional fields, but this is not explicitly documented.
  Impact: Analysts must manually check for missing data before aggregation to avoid incorrectly assuming complete records.

3.9 Agent Outlier

- The data for Agent 9 (Daffy Dumbwit) seemed inconsistent with the rest of the agents. The salary was uncommonly low (50) and the address and phone number were also strange. When checking

engagements, Agent 9 also had no engagements, so to keep the data clean and relevant, we deleted this agent from the data to focus on actionable insights for the data.

- In the future, TuneWorks can keep their data clean and up-to-date by specifying if this agent is a new hire or intern and keeping their data accurate

Summary

Overall, the dataset is structurally sound and sufficiently complete for the analysis performed, but several fields require careful handling—especially agent information, timestamp calculations, outlier durations, and inconsistent formatting of contact and address data. Proper interpretation depended on combining date/time columns, validating revenue-related fields, and reviewing outliers prior to visualization. These data quality notes help ensure the findings in the subsequent sections are both accurate and transparent.

SQL Analysis & Visual Insights

4.1 Entertainer Performance Analysis

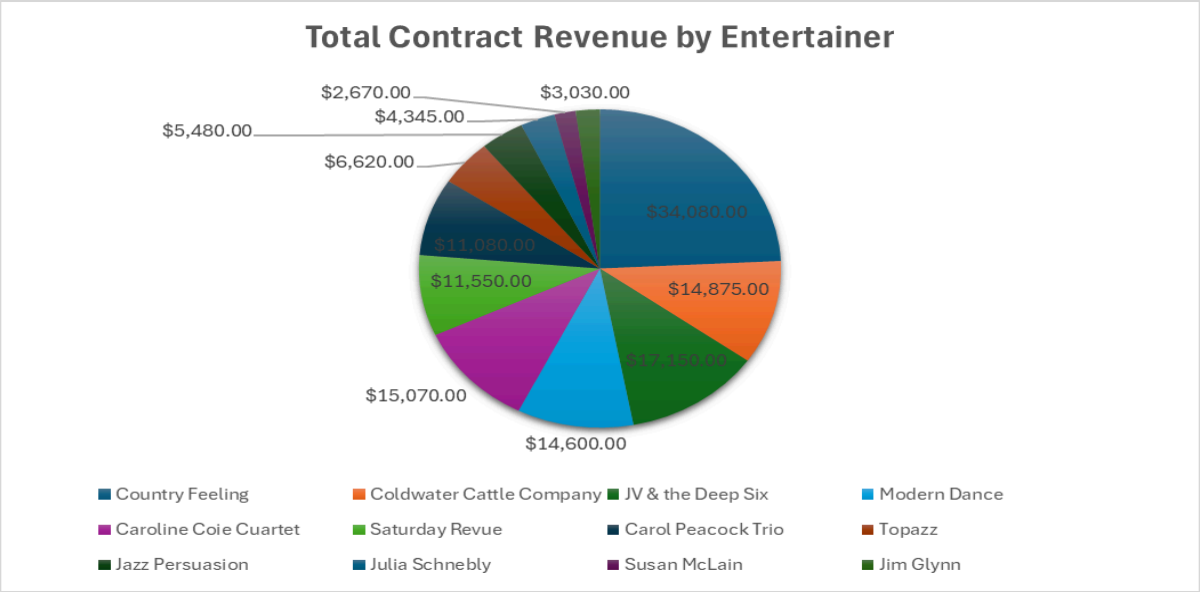
Purpose

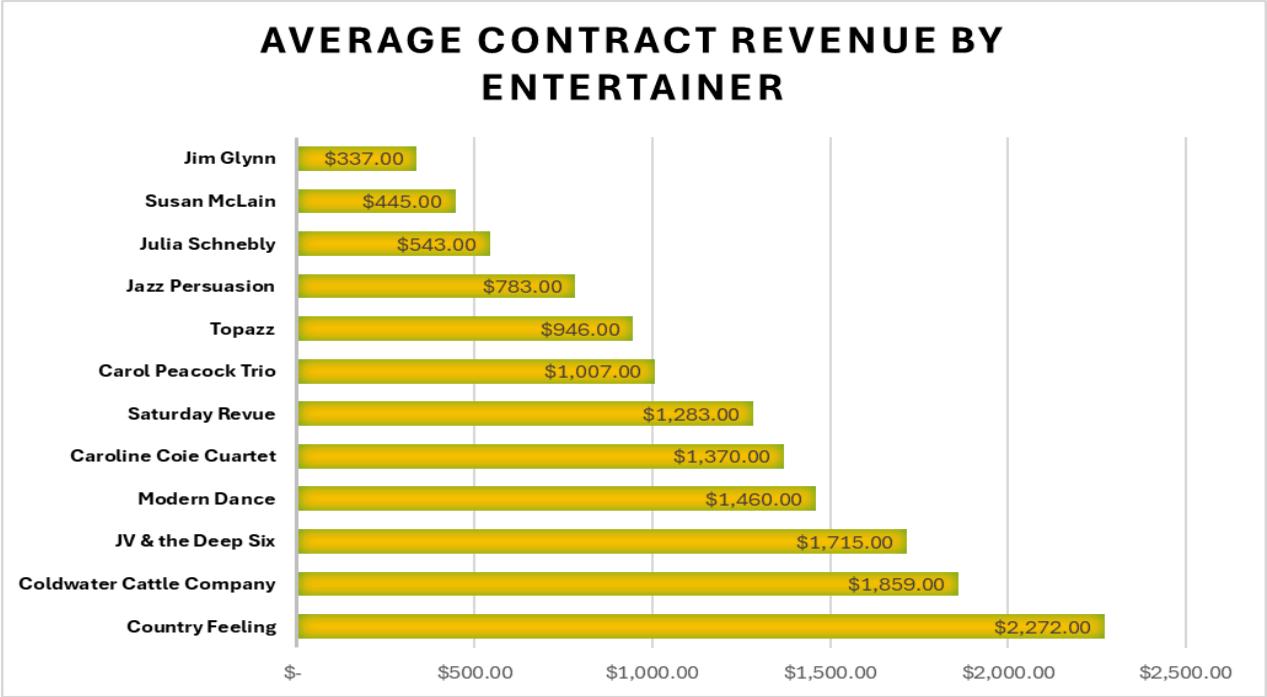
Identify which entertainers generate the most engagements, total revenue, and average revenue, in order to determine popularity and demand.

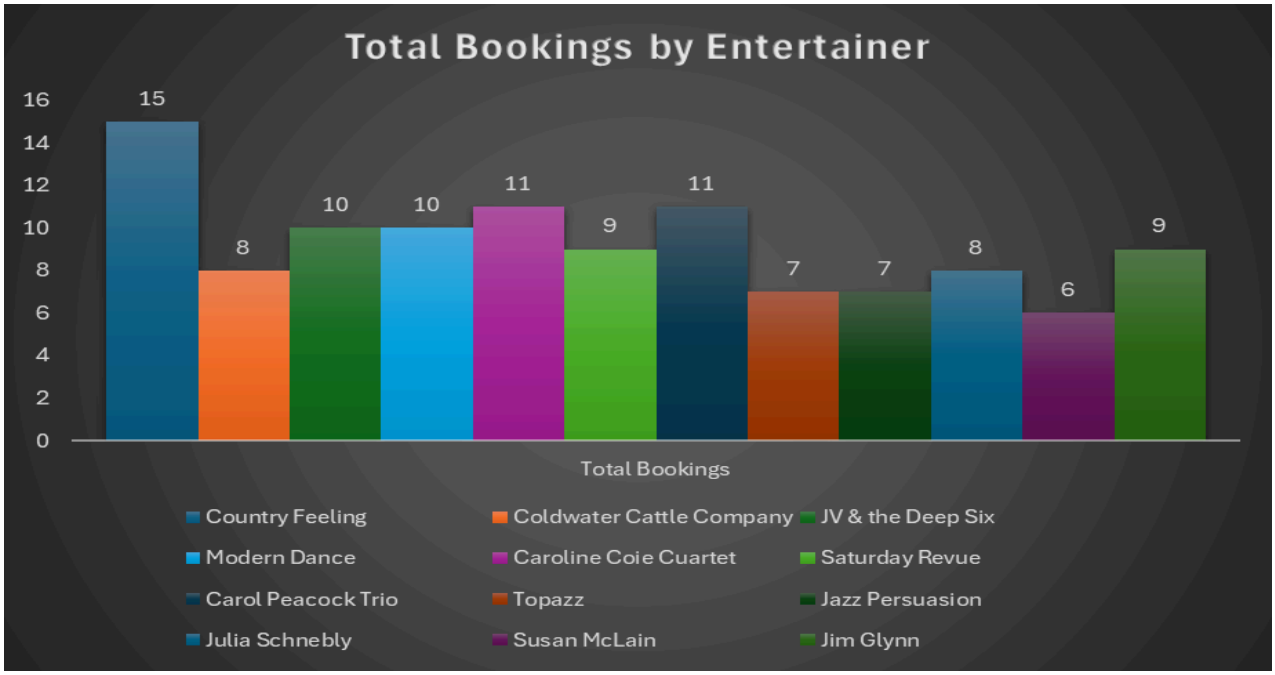
SQL Queries

```
select entertainerid, count(*) num_of_engagement, sum(e.contractprice) total_rev,
round(avg(e.contractprice)) avg_rev from engagements e
group by e.entertainerid
order by total_rev
```

Visualization







Business Interpretation

Ranking entertainers by number of engagements, total revenue, and average contract price to see who is most in demand and most valuable financially. Entertainer 1008 dominates all three metrics, clearly identifying them as a star performer. 1008 is a premium entertainer, instead of just discounting, might test premium pricing, priority booking, or bundled offers to maximize revenue while keeping demand. For low-revenue entertainers like 1012, 1004, and 1011, the data suggests a need to investigate why (e.g., low demand, poor fit, pricing issues) and decide whether to reposition, support, or eventually phase out.

4.2 Customer Value Analysis

Purpose

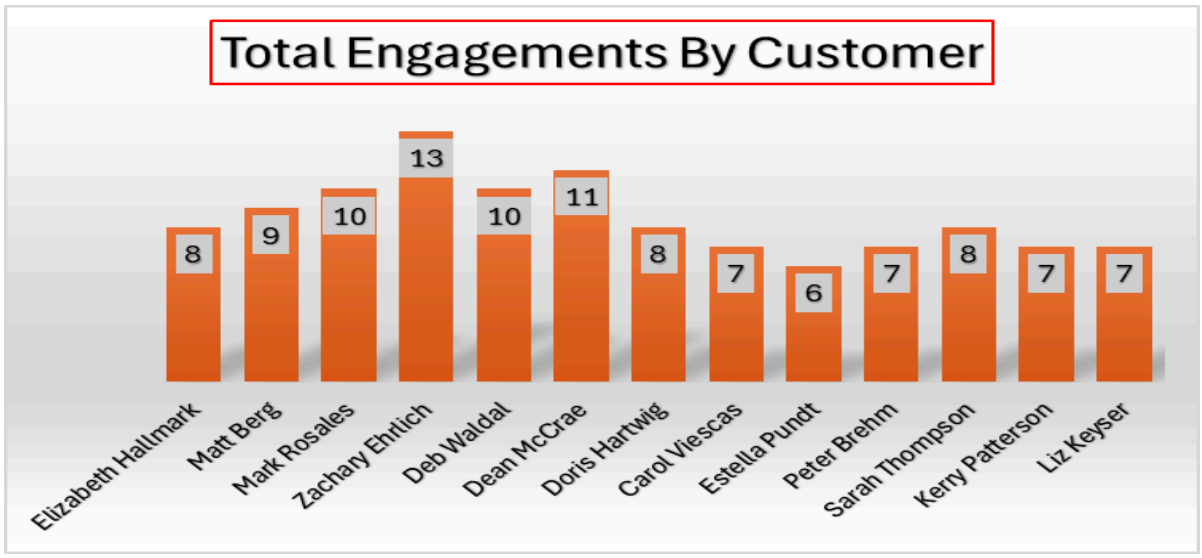
Determine which customers spend the most with TuneWorks and evaluate the revenue contribution from top clients.

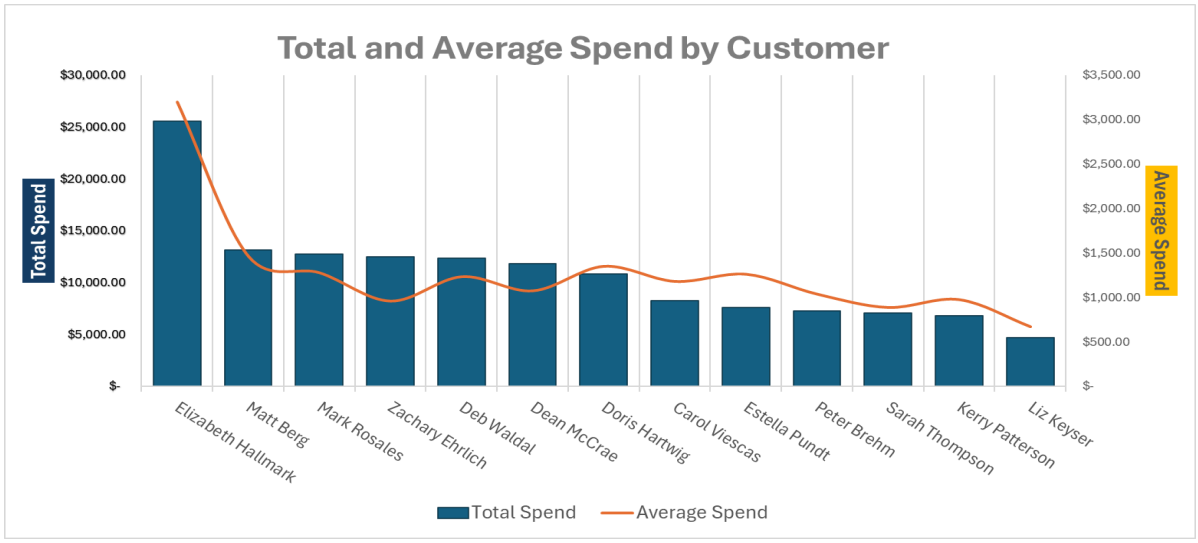
SQL Queries

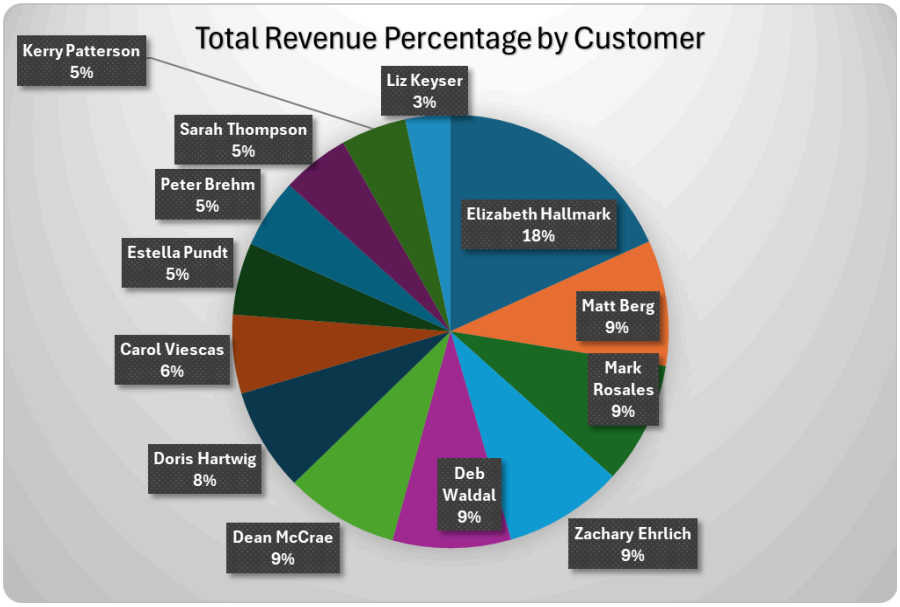
```
select e.customerid, count(*) total_engagements, sum(e.contractprice) total_spend,
round(avg(e.contractprice)) avg_spend from engagements e
group by customerid
order by total_spend desc

with customer_rev as (
 select
    e.customerid,
    sum(e.contractprice) as total_rev
 from engagements e
 group by e.customerid
),
rev_with_pct as (
 select
    cr.customerid,
    cr.total_rev,
    sum(cr.total_rev) over () as grand_total
 from customer_rev cr
)
select
 rwp.customerid,
 c.custfirstname || ' ' || c.custlastname as customer_name,
 rwp.total_rev,
 round(100 * rwp.total_rev / rwp.grand_total, 2) as pct_of_total_revenue
from rev_with_pct rwp
join customers c
 on rwp.customerid = c.customerid
order by rwp.total_rev desc
```

Visualization







Business Interpretation

This query identifies the customers with the highest total spend, along with how often they book and their average spend per engagement. Using total_spend to define “most valuable,” and customer 10005 emerges as the top spender, which are the VIP clients. They are ideal targets for loyalty programs, early access to top entertainers, or personalized service. TuneWorks should treat them differently (e.g., relationship management, occasional discounts, or recognition) to protect and grow this core revenue base. Calculates each customer’s total revenue and then computes the cumulative share of revenue as you move from the highest-spending customer downwards. It shows how concentrated revenue is, e.g., whether a small number of customers account for a large percentage of total revenue. This is a risk and strategy metric: high concentration means TuneWorks is vulnerable if a few big customers leave, but also that targeted retention of those customers is extremely valuable. It supports decisions like prioritizing key accounts, diversifying the customer base, or building backup relationships.

4.3 Monthly Engagement Trends (Seasonality)

Purpose

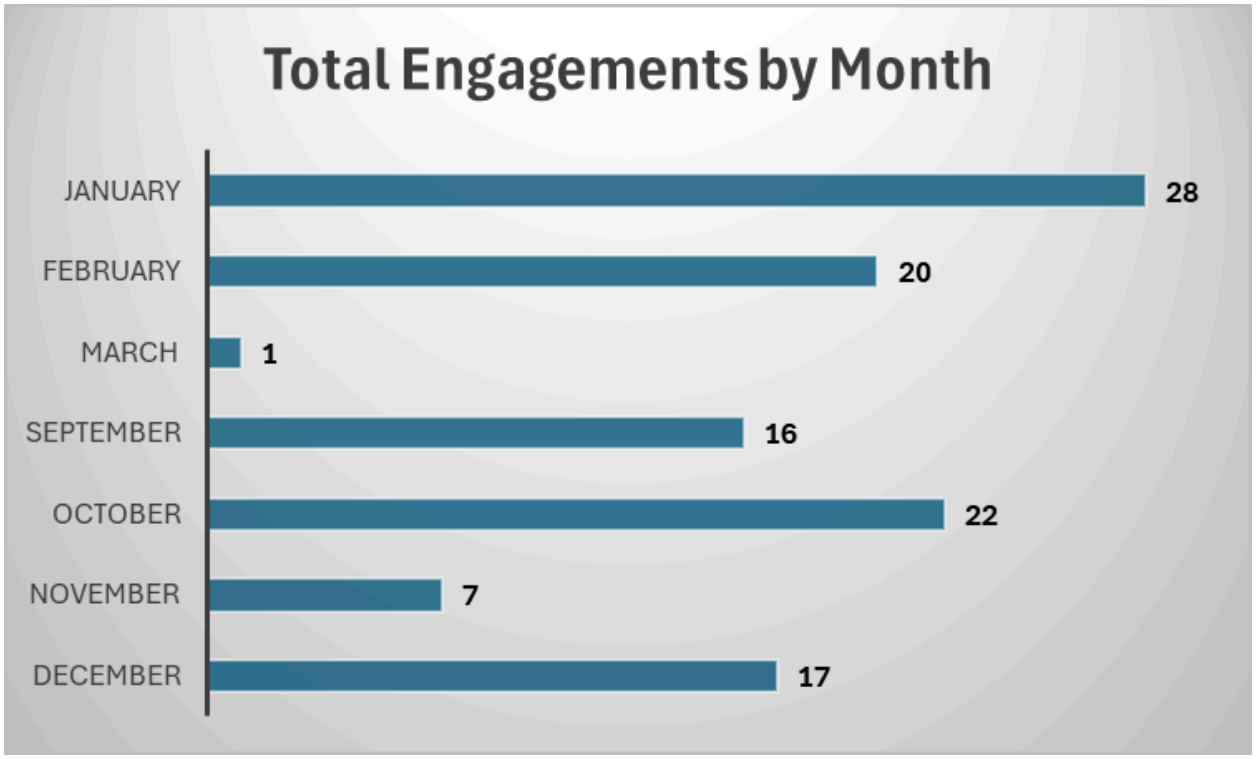
Identify which months have the highest or lowest number of engagements to assess seasonality and demand cycles.

SQL Queries

```
select
 case
```

```
    when extract(month from startdate) = 1 then 'January'
    when extract(month from startdate) = 2 then 'February'
    when extract(month from startdate) = 3 then 'March'
    when extract(month from startdate) = 4 then 'April'
    when extract(month from startdate) = 5 then 'May'
    when extract(month from startdate) = 6 then 'June'
    when extract(month from startdate) = 7 then 'July'
    when extract(month from startdate) = 8 then 'August'
    when extract(month from startdate) = 9 then 'September'
    when extract(month from startdate) = 10 then 'October'
    when extract(month from startdate) = 11 then 'November'
    when extract(month from startdate) = 12 then 'December'
 end as month_name,
 count(*) as total_engagement
from engagements e
group by month_name
order by total_engagement desc
```

Visualization



Business Interpretation

Counts engagements by month, revealing the seasonality of demand. January has the highest number of engagements while March is extremely low, with only one engagement. This suggests strong peak-season demand in January and potentially structural or market issues in March. Instead of immediately "shutting down March," TuneWorks could explore why March is weak (marketing, holidays, pricing, event cycles) and decide whether to boost demand or purposely scale back operations during that month.

4.4 Customer Segmentation by Revenue Tiers

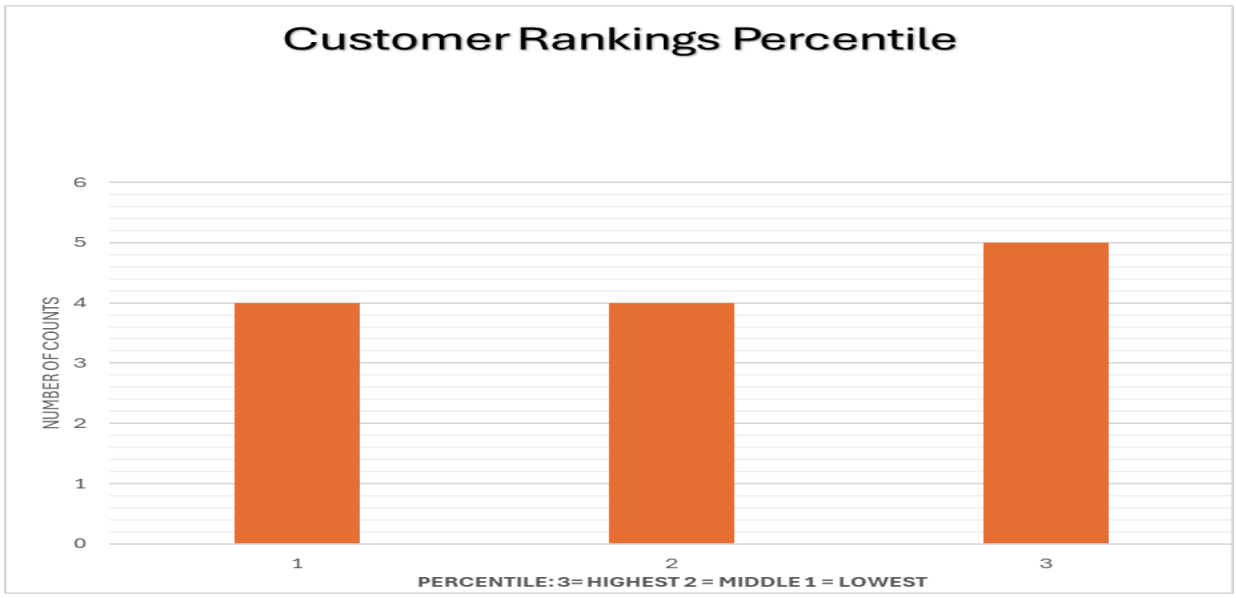
Purpose

Group customers into high, mid, and low revenue tiers to create a segmentation model for targeted strategy.

SQL Queries

```
with customer_rev as
(select customerid, sum(e.contractprice) as total_rev from engagements e
group by customerid
)
select customerid, total_rev, ntile(3) over(order by total_rev) as rev_rank from
customer_rev
order by total_rev desc
```

Visualization



Business Interpretation

Segmenting customers into three revenue tiers using NTILE(3), based on their total revenue: top, middle, and bottom groups. Top-tier customers can get VIP treatment, middle-tier customers can be nudged upward, and low-tier customers might be served more efficiently or targeted with more automated marketing. It ensures TuneWorks focuses its limited resources where the ROI is highest.

4.5 Relationship Between Duration and Revenue

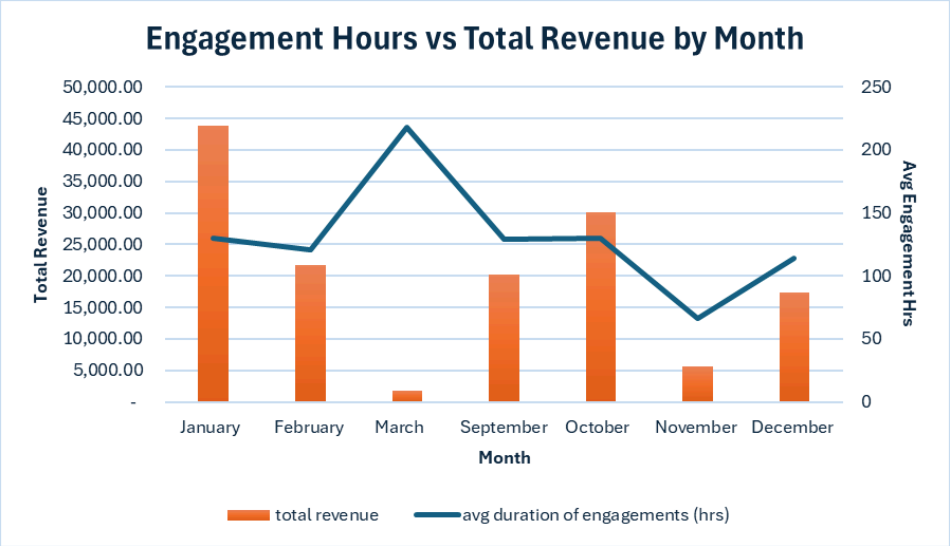
Purpose

Determine whether longer engagements generate higher total revenue.

SQL Queries

```
select extract(month from startdate) as month,
round(avg(extract(epoch from ((enddate + stoptime) - (startdate + starttime))) / 3600)) as
avg_duration_hours,
round(sum(contractprice)) as total_rev from engagements e
group by extract(month from startdate)
order by total_rev desc
```

Visualization



Business Interpretation

Calculates the average engagement duration (in hours) and the total revenue by month. Compare whether higher average durations correspond to higher total revenue and notice an outlier (March) where one 218-hour engagement skews the results. Overall, longer engagements tend to correlate with higher revenue, but the relationship must be checked for outliers before making pricing conclusions. This suggests TuneWorks could experiment with duration-based pricing or packages, while being careful about extreme cases that distort the averages.

4.6 Customer Booking Frequency (Days Between Engagements)

Purpose

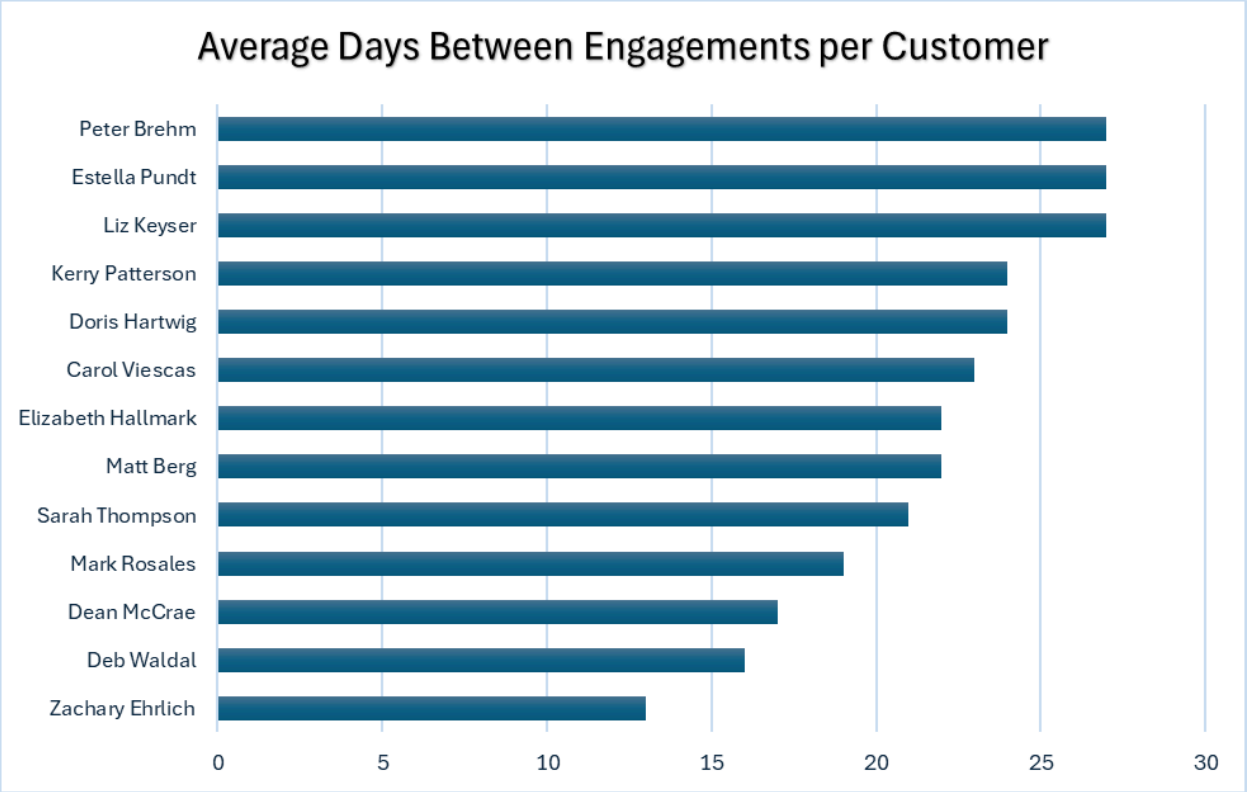
Understand how often customers return by computing their average days between bookings.

SQL Queries

```
with sorted_engagements as
(select customerid, startdate, lag(startdate)
over (partition by customerid order by startdate) as previous_engagement_date from
engagements)
select customerid, round(avg(startdate - previous_engagement_date)) as
avg_days_between_engagements from sorted_engagements
where previous_engagement_date is not null
group by customerid
```

order by ***avg_days_between_engagements***

Visualization



Business Interpretation

Calculates the average number of days between bookings for each customer, essentially their booking frequency cycle. Observed that some customers return as frequently as about every 13 days, while others have much longer gaps like 27 days or more. This is a powerful indicator of customer engagement and loyalty: shorter gaps mean highly active, recurring customers who should be nurtured. TuneWorks can use this insight to identify high-frequency customers for VIP programs and set up re-engagement campaigns when typical time-between-engagement thresholds are exceeded.

4.7 Agent productivity: Total revenue, number of engagements, and average revenue per agent

Purpose

Identify which agents generate the most engagements, total revenue, and average revenue, in order to determine productivity and workload distribution. We also analyze the spread of entertainers, musical styles, and contract prices from engagements to determine which agents manage the highest and lowest-earning entertainers.

SQL Queries

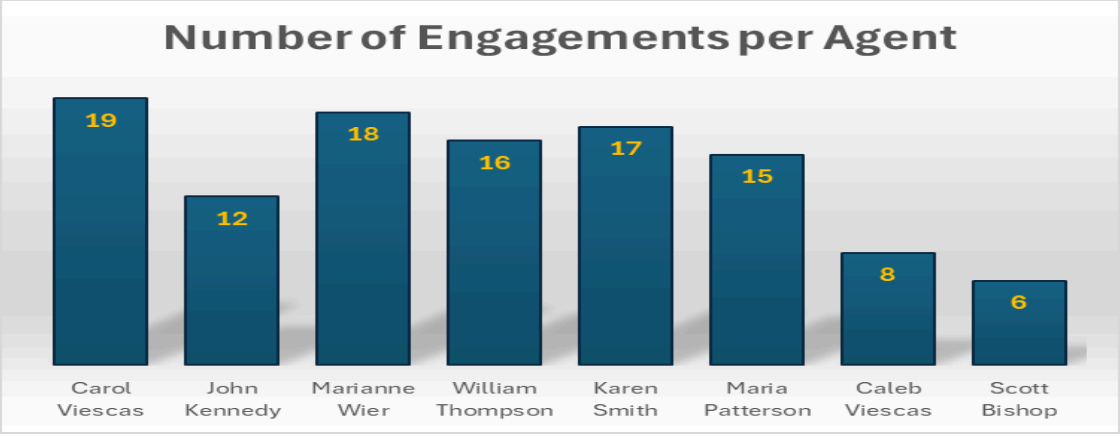
Total revenue, engagements, and average revenue per agent:

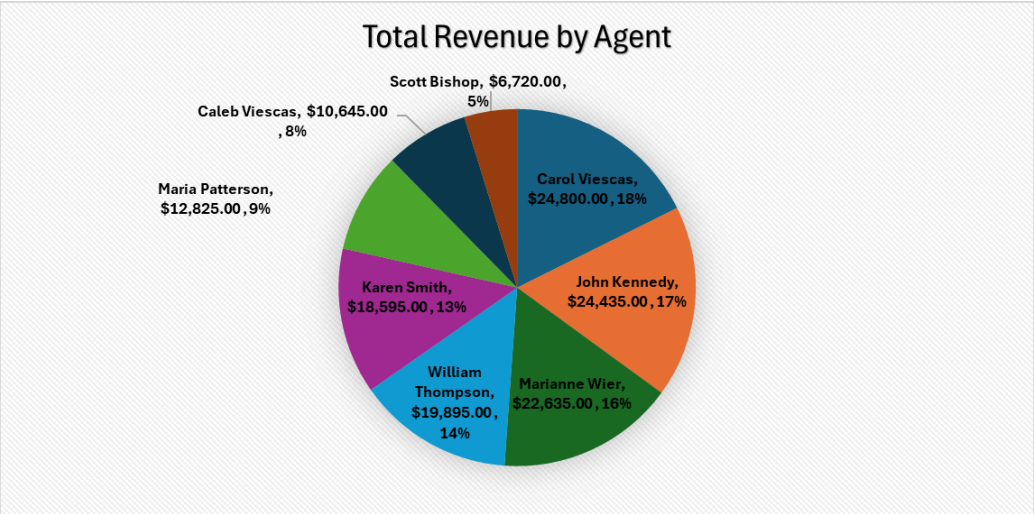
```
select a.agentid, a.agtfirstname, a.agtlastname, count(e.engagementnumber) num_engagements,
sum(e.contractprice) total_revenue, avg(e.contractprice) avg_rev_per_engagement
from agents a
left join engagements e
on a.agentid = e.agentid
group by a.agentid, a.agtfirstname, a.agtlastname
order by total_revenue desc
```

Entertainers, musical styles, and contract price (desc) for each agent

```
select eg.agentid, eg.entertainerid, m.stylename, eg.contractprice
from engagements eg
left join entertainer_styles es
on eg.entertainerid = es.entertainerid
left join musical_styles m
on es.styleid = m.styleid
where es.stylestrength = 1
order by agentid, contractprice desc
```

Visualizations





Business Interpretation

There seems to be an uneven spread of engagements and revenue earned among agents. We recommend that TuneWorks redistribute high and low-earning entertainers among agents to balance the workload and increase overall bookings, and raise agents' commission rates as an incentive to manage more bookings. This way, TuneWorks can optimize workload and increase productivity, motivation, bookings, and revenue.

Folk music, specifically the entertainer Jim Glynn, earns the lowest revenue while country music, specifically the entertainer Country Feeling, earns the highest.

Karen Smith (Agent 4) and Maria Patterson (Agent 8) have a high number of engagements to manage with low revenue per engagement, due to managing a higher number of folk music engagements. Caleb Viescas (Agent 7) had only 8 engagements but managed a higher number of country music engagements, resulting in 2nd highest average revenue. We recommend that TuneWorks redistribute folk and country music engagements, as well as increasing Caleb's commission rate as motivation to manage more bookings, since he currently has the lowest commission rate at 3.5%. This would increase productivity and bookings among agents and result in increased revenue.

Carol Viescas (Agent 3) achieved top engagements and top total revenue and has the same salary ($30k) as Maria Patterson (Agent 8), but contributes double the revenue. We would recommend a salary increase as further incentive to continue her top performance.

John Kennedy (Agent 6) achieved top revenue per engagement but has a lower salary than William Thompson (Agent 1), who has the highest salary ($35k), started working at TuneWorks

at the same time, and contributes 3% lower revenue. We would recommend a salary increase to match William Thompson at $35k as recognition of his performance.

For lower-performing agents like Scott Bishop (Agent 2), the data shows he has the lowest number of engagements and lowest percentage of revenue contributed. This suggests lack of motivation and highlights opportunities for coaching, sharing best practices from top agents like Carol and John, or rebalancing client assignments.

Conclusion

This analysis of the TuneWorks database provides a clear picture of how the company is performing across entertainers, customers, agents, and overall engagement activity. While the dataset contains some quality issues—such as messy timestamps, irregular IDs, and a few outlier engagement durations—it's still strong enough to uncover meaningful trends that inform revenue drivers, customer behavior, talent strategy, and agent performance.

Entertainer performance is highly concentrated. A small group of entertainers, most notably Entertainer 1008, generates a disproportionately large share of total revenue and bookings. This "star-power" effect suggests an opportunity for TuneWorks to revisit pricing, invest more intentionally in top performers, and better align resources to match the value those entertainers bring. On the other hand, lower-revenue entertainers may benefit from repositioning, coaching, or a strategic review of their long-term fit.

Customer behavior follows a similar pattern. A small segment of top-tier customers accounts for a large portion of total spend, with higher average bookings and higher per-engagement value. This creates both upside and risk: strong relationships with these customers are essential, but TuneWorks should also diversify its customer base to avoid over-reliance. Tiered segmentation, targeted retention efforts, and customized service bundles could help strengthen loyalty while broadening reach.

Seasonality also plays an important role. January consistently shows the highest demand, while March dips unusually low due in part to an outlier. Understanding these trends allows TuneWorks to better plan staffing, promotions, and marketing campaigns. Engagement analytics also show that longer events generally generate more revenue, though extreme outliers highlight the need for continued data cleanup.

Agent performance shows meaningful gaps as well. Top performers—such as agent Carol Viescas—regularly handle higher-value contracts and generate more revenue overall. Others manage fewer or lower-value engagements. Adjusting commission structures, balancing entertainer assignments, and rewarding strong performance can help TuneWorks increase productivity while supporting agents who may need development.

Overall, the insights suggest four main opportunities for TuneWorks:

1. Strengthen relationships with top-tier entertainers and customers while reducing concentration risk.

2. Optimize agent incentives and rebalance workloads to improve performance and retention.

3. Use seasonality trends to fine-tune marketing, staffing, and operational planning.

4. Improve data quality to support more accurate forecasting and better long-term decision-making.

By applying these data-driven recommendations, TuneWorks can boost efficiency, reduce revenue volatility, and ultimately enhance both profitability and operational effectiveness.

Key Takeaways

What did ChatGPT do better?

ChatGPT has proven itself to be a marvel of technology, and this remained true for this project as well. It was able to generate dictionaries, insights, and code instantly. ChatGPT helped us generate ideas quickly, summarize large amounts of data in seconds, and saved us a lot of time on tasks that would have taken much longer to do manually. It also made data cleaning much faster and efficient, identifying more aspects such as Missing values, Duplicates, Outliers, Invalid foreign keys / mismatched relationships, Inconsistent formats, Business-rule violations, PII concerns, and Data integrity in time dimensions.

With that being said, what we found most valuable, aside from its convenience, was the depth of its insights. It generated over 30 queries to support our EDA and even grouped those queries and the information they produced into separate themes. This gave us options when deciding the scope of our study and presented us with ideas and perspectives that we had not considered. It was also able to identify potential issues that could arise in the future.

What did your team do better?

Our team's strengths showed why human ingenuity is still more valuable than artificial intelligence. One of the primary issues we experienced with ChatGPT was that the scope of its study was sometimes too broad. There were moments when ChatGPT explored information that felt irrelevant and did not meaningfully support our main analysis, which occasionally distracted us from the primary focuses of our study. As a team, we were better at getting straight to the important parts of the data and focusing on the specific insights that were most valuable. This made our process simpler and demonstrated the value of human perspective.

We also found that ChatGPT made mistakes in cleaning and analyzing the data and offering recommendations. For example, in data cleaning, ChatGPT missed filtering out the members' genders, which is actually a null value. In recommendation 9, ChatGPT also recommended clustering customer types including venues, corporates, or individuals. However, this data is not part of our dataset and customers are only on the individual level.

What did you learn by combining both approaches?

We were able to step in and correct issues in ChatGPT's analyses and recommendations, which not only improved the insights that could benefit TuneWorks but also demonstrated the unique problem solving abilities that we have as humans.

We also learned that ChatGPT is great for speed, but judgement is still needed. Therefore, our final results are based on our team's decision and ChatGPT's ideas after checking for accuracy and relevance. We learned that combining our own work with ChatGPT allows us to be more efficient in our data cleaning, analyses, and recommendations.

What chat did better?
ChatGPT helped us generate ideas quickly, summarize large amounts of data in seconds, and saved us a lot of time on tasks that would have taken much longer to do manually.
Data cleaning: identify more aspects:such as **Missing values, Duplicates, Outliers, Invalid foreign keys / mismatched relationships, Inconsistent formats**

Business-rule violations

PII concerns

Data integrity in time dimensions

What we did better?

Data cleaning: chatgpt Missed members- genders

for the duplicate records, there are no true duplicate records in the SQL file (Customers, Agents, Entertainers, Engagements, Preferences, Members, Styles, etc.), it still generate queries for that

What did you learn by combining both approaches?

- ChatGPT is great for speed, but judgement is still needed.
- The final result is based on our team's decision and ChatGPT's support.
- ChatGPT gives us ideas, and we check accuracy and relevance.
- Combining both makes the analysis more efficient.

⭐ RECOMMENDATION 2

Create a Style-Based Marketing Strategy Targeting Customer Preferences

Rationale: Customers often book entertainers matching their preferred styles.
Impact: Higher conversion rates + increased customer satisfaction.
Data Required: Customer style preferences, booking histories.
Implementation:

- Map customers to entertainers matching preferred styles
- Automate email campaigns around preferred genres

Incorrect:⭐ RECOMMENDATION 9

Build a Customer Segmentation Framework

Rationale: Different customer types (corporate, individuals, venues) show different patterns.
Impact: Personalized offers and targeted upselling.
Data Required: Customer industry, city, past bookings.
Implementation:

- Cluster customers by revenue level, location, and style preference

- Create targeted packages